

Chapter 1

Project Objective and Business Problem

Sales forecasting is a critical business activity because organizations need to estimate future product demand and sales trends. Accurate sales prediction helps companies optimize inventory management, improve budgeting processes, allocate resources efficiently, and minimize financial losses. Forecasting helps businesses avoid problems such as overstocking and stock shortages.

The business problem addressed in this project is the prediction of future sales values using historical transaction records. Since customer demand changes because of multiple factors such as seasonality, discounts, product categories, and market behavior, predicting future sales becomes challenging.

Project objectives:

- Predict future sales using historical data.
- Forecast next month and next three-month trends.
- Compare baseline and advanced machine learning models.
- Generate useful business insights.
- Improve model performance through optimization.

Chapter 2

Dataset Description

The dataset contains historical sales transactions including customer, product, and sales-related information.

The following variables were used:

- Order Date
- Ship Date
- Product Category
- Product Name
- Quantity
- Discount
- Profit
- Shipping Cost
- Region
- Market
- Order Priority
- Sales

Dataset preparation involved:

- Detecting missing values
- Handling duplicate records
- Converting date columns
- Detecting outliers
- Transforming skewed data

Chapter 3

Data Cleaning

Raw datasets generally contain inconsistencies that negatively affect machine learning performance. Therefore data cleaning was performed before model implementation.

3.1 Missing Values

Missing values were detected using:

```
df.isnull().sum()
```

Seven missing values were found in product names. Missing values were replaced using mode imputation because the number of missing records was small.

3.2 Duplicate Records

Duplicate rows were checked and removed to avoid repeated information influencing predictions.

3.3 Outlier Detection

Boxplots and IQR methods were used for outlier detection.

Large outliers existed within sales values. Since extreme values may distort predictions, logarithmic transformation was applied.

Chapter 4

Exploratory Data Analysis

Exploratory Data Analysis was performed to understand relationships and trends.

EDA objectives:

- Understand sales distribution
- Identify trends
- Detect anomalies
- Analyze feature relationships

Key findings:

- Sales distribution was positively skewed.
- Product categories influenced sales significantly.
- Quantity showed strong influence on sales.
- Discount values affected purchasing behavior.
- Sales contained significant fluctuations.

Chapter 5

Feature Engineering and Model Selection

Feature engineering was applied to improve prediction performance.

New features extracted:

- Month
- Day
- Weekday
- Quarter
- Year

Categorical features were converted into numerical form using One-Hot Encoding.

Initially encoding created a large number of columns. Feature selection techniques were therefore applied to remove unnecessary features.

Models implemented:

5.1 Multiple Linear Regression

Multiple Linear Regression was selected as a baseline model due to its simplicity.

5.2 Random Forest

Random Forest was selected because it captures nonlinear relationships and feature interactions.

Chapter 6

Model Performance Evaluation

The following evaluation metrics were used:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 Score

Linear Regression Results:

- R^2 Score = 0.76
- RMSE = 230.88

Random Forest Results:

- Base Model R^2 Score = 0.8734
- Tuned Model R^2 Score = 0.8736

Observation:

Random Forest performed better than Linear Regression because it captured complex relationships more effectively.

Chapter 7

Hyperparameter Tuning

GridSearchCV was applied to improve model performance.

Parameters tested:

- Number of estimators
- Maximum depth
- Minimum samples split
- Minimum samples leaf

Hyperparameter tuning slightly improved prediction performance.

Chapter 8

Business Insights and Recommendations

Important business insights:

- Product categories strongly influence sales.
- Quantity affects demand significantly.
- Discounts influence customer behavior.
- Sales fluctuate across time.

Recommendations:

- Maintain inventory for highly demanded products.
- Monitor influential product categories.
- Optimize discount strategies.
- Use forecasts for resource planning.
- Improve operational decisions using predicted demand.

Chapter 9

Challenges Faced

Challenges encountered during implementation:

- Missing values in product names
- Large number of encoded columns
- Presence of outliers
- Future input generation difficulty
- Long GridSearchCV execution time

Solutions implemented:

- Mode imputation
- Feature selection
- Log transformation
- Historical trend-based future generation
- Reduced parameter combinations

Chapter 10

Future Improvements

Possible future improvements:

- ARIMA implementation
- LSTM implementation
- Holiday indicators
- External economic variables
- Real-time deployment

Chapter 11

Conclusion

This project successfully implemented a complete machine learning workflow for sales forecasting. Data cleaning, feature engineering, model building, model evaluation, and forecasting techniques were implemented successfully.

Random Forest produced stronger prediction performance compared to Multiple Linear Regression and achieved better forecasting capability.